

Performance Analysis of a SI Engine under Different Operating Conditions

Core Mechanical Engineering Project — Modeled Engineering Study

Objective

Evaluate the performance of a four-cylinder, four-stroke, naturally aspirated spark-ignition engine over 1500–5000 rpm using standard engine-performance equations and representative operating data.

Important data-use note: The operating dataset is representative/modelled data for portfolio analysis. It must not be presented as physical dynamometer measurements unless an actual experiment is performed.

1. Engine Specifications

Parameter	Value
Engine type	4-stroke, naturally aspirated SI
Cylinders	4
Bore × stroke	80 × 90 mm
Compression ratio	9.5:1
Displacement	1.8096 L
Fuel	Petrol
Calorific value	44 MJ/kg
Ambient temperature	25 °C
Ambient pressure	101.325 kPa
Air density	1.184 kg/m³

2. Calculation Method

Brake Power = $2\pi NT/60000$. Fuel flow = Brake Power × BSFC / 1000. Air flow = Fuel flow × AFR. Brake thermal efficiency is calculated from brake power and fuel energy input. Volumetric efficiency compares actual intake air volume flow with the theoretical four-stroke displacement flow rate.

3. Performance Results

Parameter	Best / Peak value	Operating point
Maximum torque	100 N·m	3500 rpm
Maximum brake power	46.08 kW	5000 rpm
Minimum BSFC	265 g/kWh	3000 rpm
Maximum BTE	30.87 %	3000 rpm
Maximum volumetric efficiency	62.59 %	4000 rpm

4. Engineering Interpretation

Torque rises to a peak of 100 N·m at 3500 rpm and then declines. Brake power continues increasing to 46.08 kW at 5000 rpm because power depends on both torque and engine speed. BSFC reaches its minimum at 3000 rpm, corresponding to the maximum BTE of about 30.87%. Volumetric efficiency reaches approximately 62.59% at 4000 rpm in the modeled dataset.

5. Engineering Consistency Checks

Displacement calculated from geometry = 1.809557 L. The MATLAB model checks displacement consistency, positive brake power, BTE limits, volumetric-efficiency limits, and positive AFR values. All modeled values satisfy these basic checks.

6. Performance Characteristic Plots





